Genetics, Epigenetics, and Gene–Environment Interplay

Chapter 9

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Central question. How much of trauma response is inherited, what does heritability actually mean, and why have simple genetic explanations failed?

Learning objectives

- distinguish genetic variation, heritability and genetic risk;
- explain why candidate-gene findings often fail to replicate;
- describe what genome-wide association studies can and cannot show;
- explain gene–environment interplay without genetic determinism;
- evaluate claims about epigenetic inheritance of trauma.

Introduction

Few areas of trauma research have produced more public misunderstanding than genetics. Headlines announce "the trauma gene", "the resilience gene" or "the gene that makes you vulnerable to PTSD". The underlying science is more interesting and much more uncertain. Post-traumatic stress disorder is polygenic, meaning that many variants of small effect contribute to risk; it is also strongly environmental, because exposure to trauma is a prerequisite; and the two interact in ways that are difficult to disentangle.

This chapter explains the concepts needed to read the literature critically. It distinguishes heritability from determinism, describes the shift from candidate genes to genome-wide studies, and examines epigenetics—the study of how gene expression is regulated without changes in DNA sequence. It closes with the ethical questions raised by genomic research on trauma.

> **A note on heritability.** *Heritability is the proportion of variation in a trait, within a particular population at a particular time, that is attributable to genetic variation. It is not the proportion of a person's condition caused by genes. A heritability of 40% does not mean that 40% of a person's PTSD is genetic and 60% environmental; it means that, in that population, 40% of the variance in the trait is associated with genetic differences.*

9.1 Genetic variation

The human genome contains about three billion base pairs. Most variation between people consists of single nucleotide polymorphisms (SNPs), common variants at a single position, together with insertions,

deletions and structural variants such as copy-number changes. Most common variants have small effects; rare variants can have larger effects and are often associated with Mendelian disorders.

Genetic variation influences trauma risk in several ways. It can affect stress physiology—for example, through variants in genes regulating the HPA axis. It can affect temperament and reactivity. It can affect the probability of exposure to certain environments, a phenomenon known as gene–environment correlation. And it can affect response to treatment.

None of these pathways is deterministic. A genetic variant changes a probability, usually slightly, in a particular context.

Definitions of the genetic terms used in this chapter follow the National Human Genome Research Institute glossary (n.d.-a).

9.2 Heritability and its limits

Twin studies compare concordance in monozygotic and dizygotic twins to estimate heritability. For PTSD, estimates cluster around 30–50%, varying with trauma type and sample. These estimates are useful for showing that genetic factors matter; they do not identify which genes, and they can be inflated by gene–environment correlation and by assumptions about equal environments.

Heritability can change with context. If everyone is exposed to the same environmental risk, genetic differences account for more of the variation in outcome; if environments vary greatly, environmental factors account for more. This is why heritability estimates from one population should not be generalised to another.

The concept of "missing heritability"—the gap between twin-study estimates and the variance explained by identified variants—remains relevant. Much of the genetic contribution to complex traits is distributed across many small effects that are difficult to detect.

9.3 Candidate gene history

Candidate-gene studies test a small number of variants chosen on theoretical grounds. The serotonin transporter variant 5-HTTLPR is the most famous example. Early studies reported an interaction with stressful life events in predicting depression; later meta-analyses found much weaker or inconsistent effects.

The candidate-gene era produced many findings that did not replicate. Reasons include small samples, flexible analysis choices, publication bias and the fact that the underlying biology is more complex than the hypotheses assumed. A variant that affects serotonin transport in vitro may have modest effects on behaviour in a person embedded in an environment.

Candidate-gene findings are not worthless—some, such as FKBP5, have replicated in multiple studies—but they should be treated as hypotheses rather than established risk factors. Clinical genetic testing for trauma vulnerability is not supported.

9.4 Genome-wide association

Genome-wide association studies (GWAS) test millions of variants for association with a trait in large samples. They require stringent correction for multiple testing and very large samples to detect small effects. For PTSD, GWAS have identified several loci, but the variance explained remains small and replication across ancestries is limited.

GWAS findings are useful for understanding biology at scale and for constructing polygenic scores, which aggregate the effects of many variants. Polygenic scores for PTSD are currently weak predictors and are not validated for clinical use. Their predictive power is lower in populations of different ancestry from the discovery sample, which raises fairness concerns.

The most important lesson from GWAS is that complex traits are polygenic and that single-gene explanations are inadequate. This is a scientific advance even though it has not yet produced clinical tools.

9.5 Gene–environment interplay

Gene–environment interaction means that the effect of an environmental exposure depends on genotype, or vice versa. Gene–environment correlation means that genetic factors influence exposure to environments—for example, through temperament, risk-taking or selection into circumstances.

Both phenomena complicate the interpretation of observational research. If a study finds that a variant moderates the effect of childhood adversity, it may be capturing interaction, correlation or confounding. Large samples, preregistered hypotheses and replication are essential.

The practical implication is that neither genes nor environments should be treated as fixed causes. Risk emerges from their interplay over development, and interventions that change environments can change outcomes regardless of genotype.

9.6 DNA methylation

DNA methylation is the addition of a methyl group to cytosine, usually in CpG dinucleotides. It typically represses transcription when it occurs in gene promoters. Methylation patterns are established during development and can be modified by experience.

In trauma research, methylation of FKBP5 and NR3C1 (the glucocorticoid receptor gene) has been studied in relation to childhood adversity. Some studies report associations with altered stress reactivity; others do not. Samples are often small, and peripheral tissue (blood, saliva) is used as a proxy for brain.

The epigenetic clock—measures of biological age based on methylation—has been used to study accelerated ageing in trauma-exposed populations. Findings are interesting but preliminary, and the clinical significance is unclear.

The description of epigenetic mechanisms follows the National Human Genome Research Institute epigenomics fact sheet (n.d.-b).

9.7 Histone modification

Histones are proteins around which DNA is wrapped. Chemical modifications—acetylation, methylation, phosphorylation—alter chromatin structure and gene accessibility. Histone acetylation generally increases transcription; certain histone methylations repress it.

Animal studies show that stress and caregiving experiences can alter histone modifications in brain regions relevant to stress regulation. In rodents, variations in maternal care are associated with lasting epigenetic differences in offspring, including in the glucocorticoid receptor gene.

Human evidence is limited by the difficulty of obtaining brain tissue. Post-mortem studies exist but are small and affected by agonal state and tissue quality. Peripheral measures are indirect. The field is promising but not yet able to support clinical claims.

9.8 Non-coding RNA

Non-coding RNAs regulate gene expression post-transcriptionally. MicroRNAs bind to messenger RNAs and inhibit translation or promote degradation. They are involved in synaptic plasticity, inflammation and stress responses.

Circulating microRNAs in blood have been studied as potential biomarkers for PTSD and other conditions. Some studies report differences between groups, but findings are inconsistent and the biological meaning of circulating microRNAs is unclear. They may reflect immune activation, tissue damage or stress-related changes in blood cells rather than brain processes.

Long non-coding RNAs are less well understood but are increasingly recognised as regulators of chromatin and transcription. Their role in trauma is largely unexplored.

9.9 Intergenerational questions

Interest in intergenerational transmission intensified after studies reported epigenetic changes in the offspring of people exposed to famine, genocide or trauma. The mechanisms proposed include altered maternal care, altered stress physiology, and direct epigenetic inheritance through gametes.

The evidence is strongest for indirect pathways: trauma affects parenting, family functioning, economic stability and exposure to adversity, all of which affect children. Direct epigenetic inheritance in humans is plausible in principle but has not been convincingly demonstrated for trauma-related outcomes.

Claims that trauma is "passed down in the genes" should therefore be translated into more accurate statements: children of trauma survivors may face higher risk through multiple pathways, including caregiving, environment and possibly biology. This framing supports support for families rather than fatalism.

9.10 Ethical genomics and precision research

Genomic research on trauma raises ethical issues. Genetic data are identifiable and can reveal information about relatives who did not consent. Findings may be misused to justify stigma, discrimination or deterministic claims about groups. Research in under-resourced settings

must involve local scientists and communities and must not extract data without capacity building.

Genetic testing for trauma vulnerability is not clinically indicated. Direct-to-consumer tests that claim to assess trauma risk are not supported by evidence. Researchers should communicate uncertainty clearly and avoid reinforcing genetic essentialism.

9.11 Model organisms and what they can show

Much of what is known about the molecular biology of stress comes from rodents. Animal models allow controlled exposure, direct tissue sampling and experimental manipulation that would be impossible in humans.

Rodent models of chronic stress—restraint, social defeat, unpredictability—produce behavioural and endocrine changes that resemble aspects of human stress-related disorders. Maternal separation and limited bedding models are used to study early adversity. These models have identified mechanisms: glucocorticoid receptor regulation, CRH expression, hippocampal plasticity, microglial activation and epigenetic modification.

The limitations are substantial. A mouse is not a person, and a laboratory stressor is not a traumatic event embedded in meaning, relationship and culture. Measures such as freezing and corticosterone are proxies, not equivalents of human symptoms. Sex differences are often ignored because early studies used male animals, and this has distorted the literature.

Findings should therefore be treated as mechanistic hypotheses to be tested in humans, not as direct evidence about human experience. When a paper reports that a stress paradigm "causes depression-like behaviour" in rodents, the phrase in quotation marks is doing important work.

9.12 Reading genetic association studies critically

Several questions help in appraising a genetic study of trauma-related outcomes.

What was the sample size, and was it adequate for the effect being sought? Small studies in this field have produced many findings that later shrank or disappeared.

Was the phenotype well defined? "PTSD" assessed by self-report questionnaire is not the same as a clinical diagnosis, and trauma exposure is often measured retrospectively.

Was the analysis preregistered? Candidate-gene studies with flexible covariates and subgroup analyses are particularly prone to false positives.

Was the finding replicated in an independent sample, and in more than one ancestry group? Polygenic scores derived from European samples often perform poorly elsewhere.

Were confounders addressed? Population stratification, gene–environment correlation and assortative mating can produce spurious associations.

Does the effect size matter? A variant that changes risk by a few per cent may be scientifically interesting and clinically useless.

A single paper rarely justifies a change in practice. The most useful genetic findings are those that replicate across consortia, converge with other lines of evidence, and illuminate a mechanism.

Synthesis

Table 9.1 summarises genetic concepts and their clinical status.

Table 9.1. Genetic concepts and clinical status

Concept	Meaning	Current status for trauma	Caution
Heritability	Variance attributable to genetic differences in a population	30–50% in twin studies	Not individual causation
Candidate gene	Hypothesis-driven variant	Mostly unreplicated	Do not test clinically
GWAS	Genome-wide association	Several loci, small effects	Limited ancestry diversity
Polygenic score	Aggregated variant effects	Weak predictor	Not clinical
Methylation	DNA modification affecting expression	Associations reported	Peripheral tissue only
Intergenerational	Transmission across generations	Indirect pathways strongest	Avoid genetic determinism

Composite case study

A patient asks whether a genetic test can explain why she developed PTSD while her sister did not. The clinician explains that PTSD is polygenic and environmentally contingent, that no test can identify a specific cause, and that differences in exposure, support and coping are likely to matter more than any single variant. The conversation addresses the underlying wish—to understand why—without offering false certainty.

Review and discussion questions

Why is heritability not a measure of individual causation?

Why did candidate-gene studies largely fail to replicate?

What can polygenic scores currently do, and what can they not do?

Distinguish gene–environment interaction from gene–environment correlation.

Why is peripheral methylation an imperfect proxy for brain methylation?

What are the ethical risks of genetic research on trauma in low-resource settings?

Research assignments

- Design a GWAS of PTSD that includes diverse ancestries and specifies replication.
- Critically appraise one study of FKBP5 methylation and childhood adversity.
- Write a plain-language explanation of heritability for a community audience.

Chapter summary

Trauma risk is influenced by many genetic variants of small effect, interacting with environment throughout development. Heritability describes population variance, not individual causation. Candidate-gene findings have mostly not replicated; GWAS has shown that PTSD is highly polygenic. Epigenetic mechanisms regulate gene expression and

are affected by experience, but human evidence, especially for intergenerational inheritance, is limited. Genomic research must be conducted and communicated with attention to uncertainty and justice.

References

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